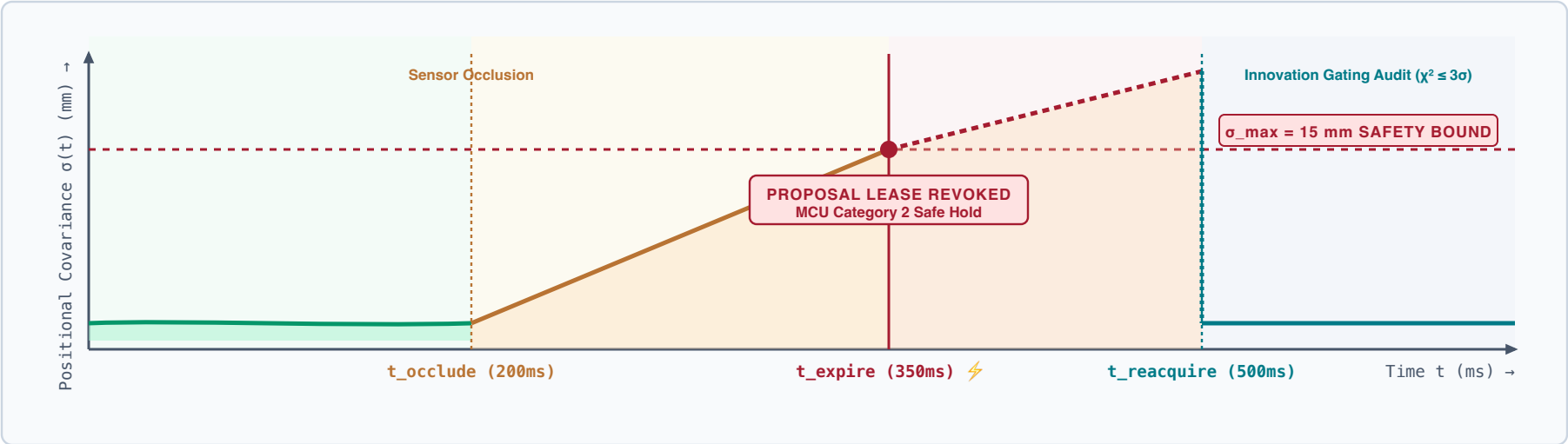


SPATIAL UNCERTAINTY EXPANSION, TTL LEASE EXPIRATION, & INNOVATION GATING

Autonomous revocation of unverified learned proposals when sensory occlusion expands spatial covariance beyond safety bounds



1. UNCERTAINTY DRIFT DYNAMICS

$$\sigma_{pos}(t) = \sigma_0 + v_{drift} \cdot \Delta t$$

- Without external vision updates, open-loop dead reckoning accumulates error.
- Target moving at $v = 0.1\text{ m/s}$ adds **10 mm error every 100 ms**.
- State covariance matrix Σ expands unboundedly.

2. HARDWARE LEASE REVOCATION

```
if (sigma_pos > sigma_max) RevokeAuthority()
```

- MCU evaluates spatial uncertainty every 1 ms tick in FreeRTOS.
- At $t_{expire} = 350\text{ ms}$, intent lease expires $\implies$ Zero motor authority.
- Executes bumpless deceleration to zero velocity (Category 2 Stop).

3. χ^2 INNOVATION GATING AUDIT

$$d_M^2 = (z - \hat{z})^T S^{-1} (z - \hat{z}) \leq \gamma$$

- Upon sensory return, Mahalanobis distance d_M audits the raw data.
- Valid ($\leq 3\sigma$):** Resets covariance; safely resumes trajectory execution.
- Outlier ($> 3\sigma$):** Rejects spurious visual hallucination or reflection.